Assignment 7: Baseline Models

CNN-Based Classification of Mars Surface Imagery Under Limited Data Conditions

Uladzislau Staliarou / s29905

1 Baseline Models

Stratified majority (trivial reference). Predicts whichever of the two largest classes; sets the absolute floor and exposes the accuracy/macro-F1 split.

Multinomial logistic regression on color descriptors (simple model). 48-bin RGB histogram plus per-channel summary stats (mean, std, min, max, 5th/95th percentiles), 66 features total. Motivated by Brigato & Iocchi [4]—linear models on hand-crafted features compete more often than expected on small data—and by the fact that the three Curiosity instruments differ measurably in white balance and exposure. Expected weakness: anything spatial.

Random forest on color + texture descriptors (classical ML). Same color features plus HOG (8×8 cells, 9 orientations) and a 26-bin uniform LBP histogram. Pre-CNN Mars work [2, 3] used tree-ensemble or shallow classifiers on hand-crafted features as their reference; this matches that pattern. Robust, no real tuning needed. Expected weakness: a fixed feature set—whatever HOG and LBP miss is gone.

2 Training Protocol

70/15/15 stratified split (deduplicated before splitting), reused across all models. 5-fold stratified CV on the training portion; test set touched once for Table 1. Images centre-cropped and resized to 224×224. Features standardised per training fold only. Augmentation off—these are not models that benefit from photometric jitter. Seeds fixed at `42`. Tuning runs on inner CV of the training fold and never touches the test set.

3 Hyperparameters

Both real models reported in *default* and *tuned* configurations. Tuning is intentionally minimal: the goal is to confirm the model is not pathologically misconfigured, not to chase

the last F1 point.

- **LogReg.** 1-D grid over $C \in \{0.01, 0.1, 1, 10\}$, 5-fold inner CV.
- **RF.** `n_estimators` $\in \{200, 500\} \times$ `max_features` $\in \{\sqrt{d}, \log_2 d\}$, with `class_weight="balanced"` fixed on. Eight configurations.

4 Results

Model	Acc.	Macro F1	CV F1 (mean±sd)	Rec. ML	Rec. MR	Rec. MH
Stratified majority	0.36	0.18	0.18 ± 0.01	0.00	0.50	0.50
LogReg (default)	0.63	0.60	0.59 ± 0.02	0.41	0.69	0.70
LogReg (tuned, C=0.1)	0.65	0.62	0.61 ± 0.02	0.45	0.70	0.71
Random Forest (default)	0.71	0.69	0.68 ± 0.02	0.55	0.74	0.78
Random Forest (tuned)	0.74	0.71	0.70 ± 0.02	0.59	0.76	0.80

Table 1: Test-set performance. CV column is 5-fold CV on the training portion. ML = Mastcam-L, MR = Mastcam-R, MH = MAHLI.

CV and test numbers stay within one standard deviation for every non-trivial model, suggesting no leakage. Tuning buys 2–3 F1 points over defaults—the right order of magnitude for a one-dimensional grid.

5 Interpretation

The color signal is real. A linear classifier on a 66-dimensional color descriptor reaches 0.62 macro F1. A non-trivial fraction of the task is solvable from white balance and exposure alone, which changes how any future CNN result should be read.

Texture helps but plateaus. Adding HOG and LBP and a non-linear model gains 9 F1 points ($0.62 \to 0.71$)—meaningful, consistent across folds, still below the 0.80 target. This is the gap a learned representation is supposed to close, matching what Wagstaff et al. [1] reported before switching to AlexNet.

Mastcam-L is the weak class. Per-class recall stays at 0.59 even after class weighting, fifteen points below the others. Underrepresented (28%) and visually close to Mastcam-R, so any model relying on global descriptors will tend to confuse them.

Justifies moving on. 0.71 is below the 0.80 target; that gap is the empirical case for trying more expressive models, with the condition that any future result lands in this same table on the same splits.

References

[1] Wagstaff, K.L., Lu, Y., Stanboli, A., Grimes, K., Gowda, T., & Padams, J. (2018). Deep Mars: CNN Classification of Mars Imagery for the PDS Imaging Atlas. *Proc. AAAI Conference on Artificial Intelligence*, 32(1).

[2] Rothrock, B., Kennedy, R., Cunningham, C., Papon, J., Heverly, M., & Ono, M. (2016). SPOC: Deep Learning-based Terrain Classification for Mars Rover Missions. *AIAA SPACE Forum*, AIAA 2016-5539.

[3] Palafox, L.F., Hamilton, C.W., Scheidt, S.P., & Alvarez, A.M. (2017). Automated detection of geological landforms on Mars using Convolutional Neural Networks. *Computers & Geosciences*, 101, 48–56.

[4] Brigato, L. & Iocchi, L. (2021). A Close Look at Deep Learning with Small Data. *Proc. International Conference on Pattern Recognition (ICPR).*